PPE Compliance Check Agent

Tier 3 Autonomous AI Agent for PPE Safety Monitoring

Eva Abou Harb
ITAI 1378 – Computer Vision
Houston City College
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Project Overview

A Tier 3 autonomous computer vision agent

Goal: Automate PPE helmet compliance monitoring in construction and industrial environments.

DETECT

YOLO11 identifies helmets and missing helmets.

REASON

The agent evaluates PPE compliance.

ACT

Generates evidence, reports, metrics, and traces.

Project Tier: Tier 3 – Custom-Trained CV Agent

Tier 3 qualification: custom-trained YOLO11 PPE detection model

Problem & Solution

PROBLEM

Manual PPE monitoring can be:

- Time-consuming
- Inconsistent
- Difficult to scale
- Dependent on continuous human observation

SOLUTION

PPE Compliance Check Agent:

- Detects helmet use automatically
- Identifies potential violations
- Handles uncertain cases
- Generates documented evidence
- Supports image and video monitoring

Computer Vision + AI Reasoning → Automated Safety Monitoring

Agent Architecture

Perception → Reasoning → Action

1. Input Ingestion
2. Input Validation
3. Preprocessing
4. YOLO11 Perception
5. Compliance Reasoning
6. Action & Evidence
7. Trace Logging
8. Evaluation
9. Continuous Monitoring

End-to-End Agent Pipeline

Dataset & YOLO11 Model

MODEL & DATA

- Custom PPE dataset
- Training, validation, and evaluation
- Custom-trained YOLO11 detector
- Helmet / no-helmet conditions
- Image and video inference
- Representative samples stored in GitHub

PPE Dataset — YOLO11 — Detector — Detection

How the Agent Works

Image / Video Input — YOLO11 Detection — Helmet / No-Helmet — Compliance Reasoning — Decision — Evidence & Reports

COMPLIANT

Required PPE detected.

VIOLATION

Missing PPE detected.

REVIEW_REQUIRED

Uncertain visual evidence.

Outputs: Annotated Images • Videos • JSON Reports • Metrics • Execution Traces

PPE Compliance Results

Three decision outcomes from the agent

COMPLIANT

Required PPE detected.

VIOLATION

Missing PPE detected.

REVIEW_REQUIRED

Insufficient or uncertain evidence.

Evaluation & Metrics

10 representative scenarios

COMPONENT-LEVEL

- Object detection performance
- Inference latency
- Detection consistency
- Annotated output quality

KEY MODEL RESULTS

Precision: **0.9147**
Recall: **0.8911**
mAP50: **0.9285**
mAP50–95: **0.6053**
Inference: **3.7 ms/image**

SYSTEM-LEVEL

- Successful image/video processing
- PPE compliance classification
- Annotated evidence generation
- Structured report generation
- Execution trace generation

AGENT EVALUATION

10 scenarios evaluated
6 COMPLIANT
2 VIOLATION
2 REVIEW_REQUIRED
100% processing success
Average agent inference time: **11.71 ms**

Artifacts: Evaluation Images • Monitoring Videos • JSON Reports • Metrics • Agent Traces

Failure Case Analysis

Observed Model Limitations

CASE 1:

Partial Detection in Crowded Scene

↓

Multiple workers and partial occlusion

↓

Some worker-level PPE detections missed

CASE 2:

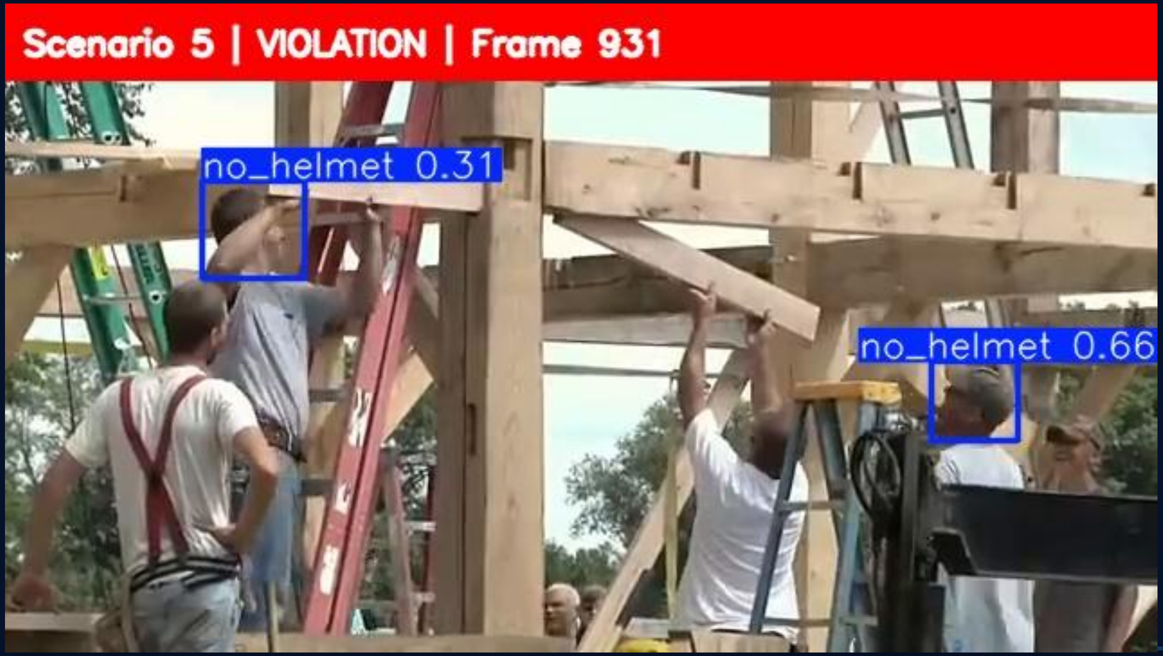
Small & Distant PPE Objects

↓

Workers occupy a small area of the frame

↓

Lower-confidence PPE detection



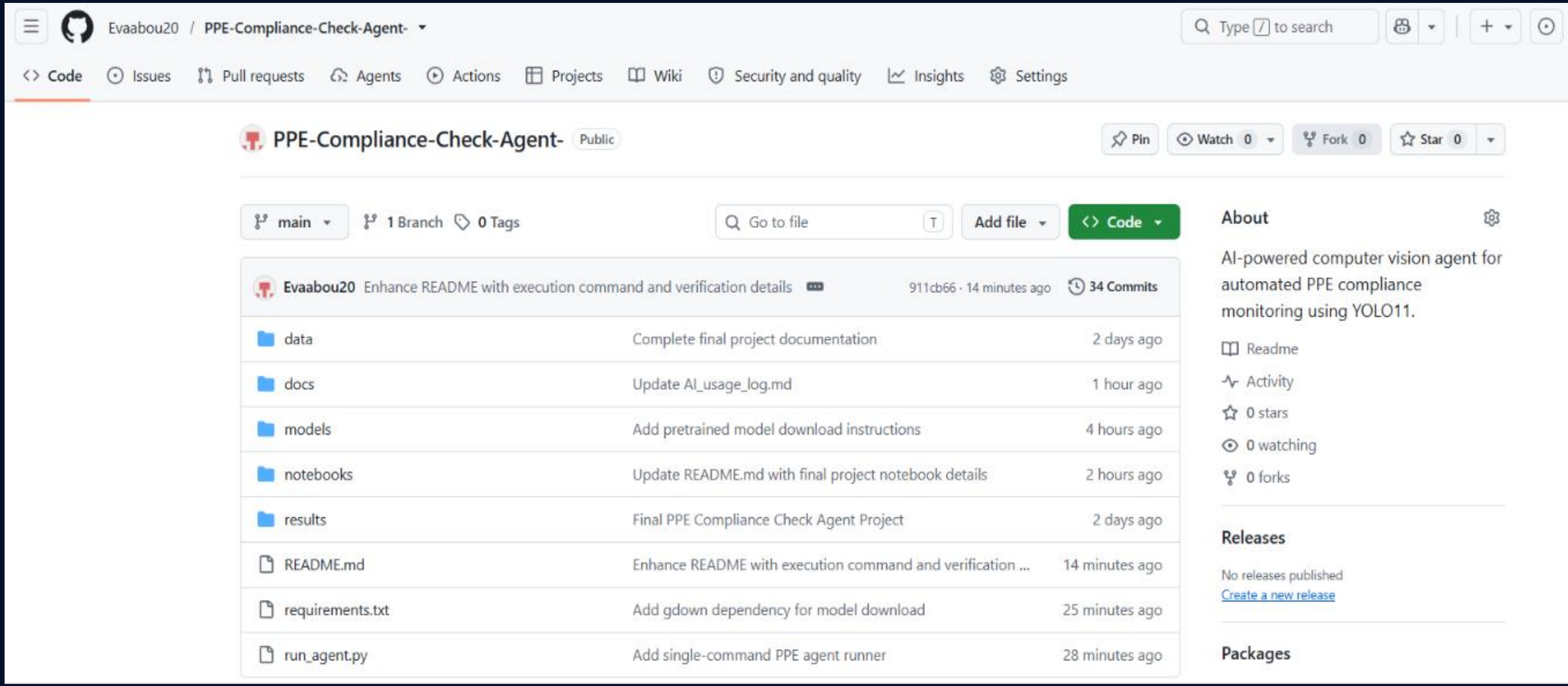
Key takeaway:

The agent completes the end-to-end workflow, but detection performance can decrease with occlusion, crowded scenes, and small or distant PPE objects.

GitHub & Reproducibility

Repository Structure

data/
docs/
models/
notebooks/
results/
README.md
requirements.txt
run_agent.py



Single-command execution: python run_agent.py
Includes: final notebook • sample data • results • metrics • reports • traces • documentation

Key Learnings & Future Improvements

KEY LEARNINGS

- Train and evaluate a custom YOLO11 detector
- Build an end-to-end computer vision agent
- Combine perception, reasoning, and action
- Evaluate AI systems systematically
- Document projects for reproducibility

FUTURE IMPROVEMENTS

- Detect vests, gloves, goggles, and boots
- Integrate live surveillance cameras
- Deploy as a real-time application
- Improve robustness to occlusion and lighting
- Expand reasoning capabilities

PPE Compliance Check Agent

Computer Vision + Reasoning + Evaluation

for Automated PPE Safety Monitoring

DETECT

Automated PPE detection with YOLO11

DECIDE

Intelligent compliance decision-making

DOCUMENT

Evidence, reports, metrics, and traces

Thank You!

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